CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A square is inscribed in a circle of radius 7 cm. The area of the circle outside the square is: 1

(a) 56 cm² **(b)** 105 cm² **(c)** 98 cm² **(d)** 154 cm²

2. The largest possible circle is cut from a square sheet of side 56 cm. The area of the sheet left over is: 1

(a) 2464 cm² **(b)** 1344 cm² **(c)** 3136 cm² **(d)** 672 cm²

3. A circular park of radius 20 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 1034 m² **(b)** 517 m² **(c)** 2068 m² **(d)** 880 m²

4. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 60° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 462 cm² **(b)** 38.5 cm² **(c)** 154 cm² **(d)** 77 cm²

5. An arc of length 22 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 45° **(b)** 180° **(c)** 90° **(d)** 120°

6. A chord of a circle of radius 49 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 1200.5 cm² **(b)** 1286.25 cm² **(c)** 1886.5 cm² **(d)** 686 cm²

7. A wire in the form of a circle of radius 21 cm is bent into a square. The area of the square is: 1

(a) 1089 cm² **(b)** 33 cm² **(c)** 1386 cm² **(d)** 4356 cm²

8. A quadrant of a circle of radius 14 cm is cut from a square of side 14 cm, with the centre of the quadrant at one corner of the square. The area of the remaining part of the square is: 1

(a) 196 cm² **(b)** 42 cm² **(c)** 154 cm² **(d)** 84 cm²

9. A square is inscribed in a circle of radius 14 cm. The area of the circle outside the square is: 1

(a) 392 cm² **(b)** 420 cm² **(c)** 616 cm² **(d)** 224 cm²

10. The largest possible circle is cut from a square sheet of side 70 cm. The area of the sheet left over is: 1

(a) 4900 cm² **(b)** 2100 cm² **(c)** 3850 cm² **(d)** 1050 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 15 minutes. 2

12. A horse is tied at the midpoint of one side of a large square field with a rope 14 m long. Find the area of the field it can graze. 2

13. A wheel has radius 35 cm. Find the number of complete revolutions it makes to cover 55 m. 2

14. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 30 minutes. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A circular field has radius 14 m. Fencing costs ₹ 12 per metre. Find the cost (in ₹) of fencing the field once. 3

16. The radii of two circles are 14 cm and 7 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

17. The radii of two circles are 3 cm and 4 cm. Find the radius of the circle whose area equals the sum of their areas. 3

OR

A circular field has radius 21 m. Fencing costs ₹ 9 per metre. Find the cost (in ₹) of fencing the field once. 3

18. The radii of two circles are 18 cm and 11 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

OR

The radii of two circles are 8 cm and 15 cm. Find the radius of the circle whose area equals the sum of their areas. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 45° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

An umbrella has 8 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 42 cm. Find the area between two consecutive ribs. 5

20. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 72° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

An umbrella has 14 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 35 cm. Find the area between two consecutive ribs. 5

END OF QUESTION PAPER — Total: 40 Marks